

Chapter 17.3

Faster RCNN Architecture

2015-2016

தமிழில்

Less than 30 Mins

AI COACH JOHN

CEO & FOUNDER

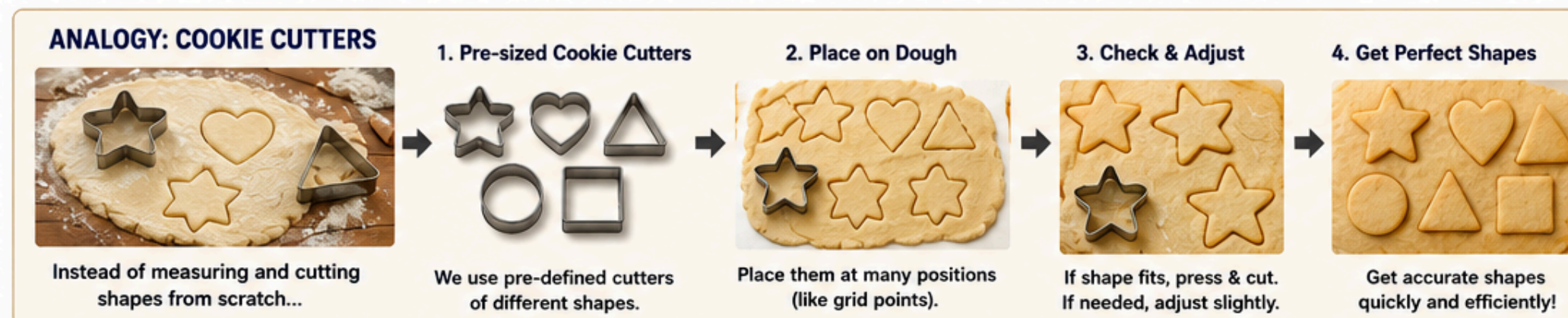
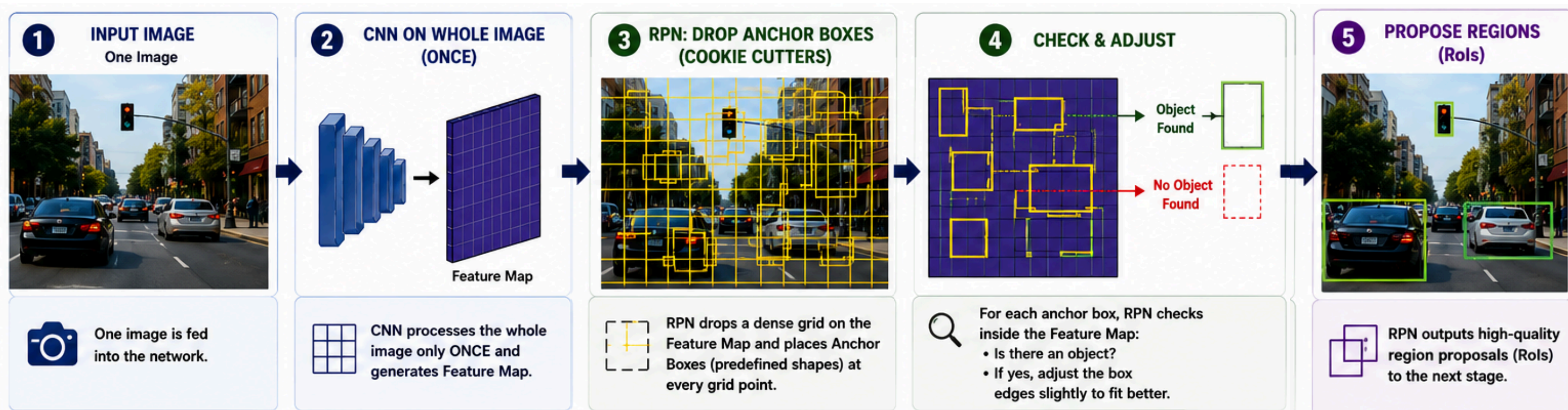
Agenda at a glance

1. Architectural Shift: **Faster RCNN**
2. Real-Time Usecases
- 3. Limitations**
4. Takeaway

Point 1

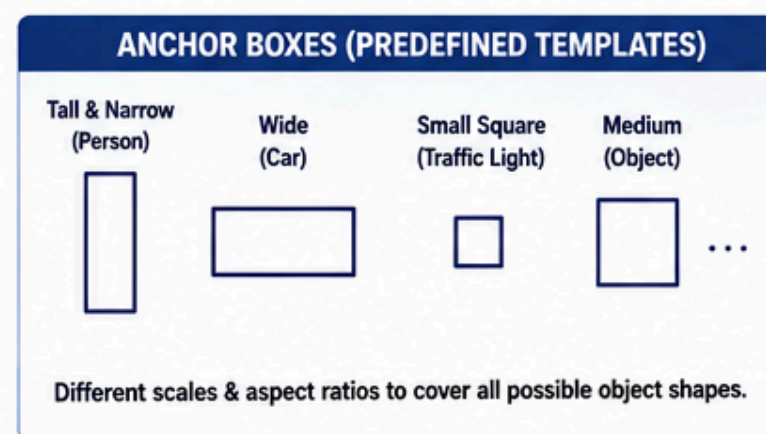
Faster R-CNN Architecture

Faster RCNN Architecture



ARE THESE BOXES PRE-DEFINED TEMPLATES?

- ✓ Yes! Anchor Boxes are pre-defined templates with different aspect ratios and sizes.
 - ✓ They act like "Cookie Cutters" for objects of various shapes.
 - ✓ GPU doesn't search from scratch; it performs fast template matching.
 - ✓ Everything runs in GPU pipeline for maximum speed.
- 



Fast RCNN → Faster RCNN

2 Sec

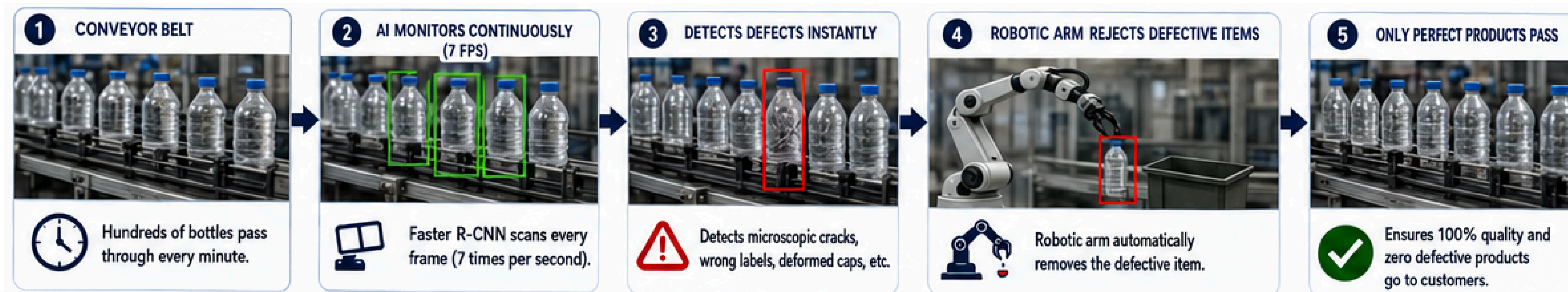
~7 FPS



Point 2

Real World Use Case

1. Manufacturing Industries → AQC(Automated Quality Control)



INDUSTRIES THAT BENEFIT

-  Beverage (Bottles, Cans)
-  Pharmaceutical (Vials, Labels)
-  Food Packaging (Seals, Labels)
-  Electronics (Soldering, Components)
-  Automotive (Parts Inspection)
- ... and many more!

Key Takeaway

1. **Predefined Anchor Boxes** (cookie cutters) enable *template matching on GPU*

This makes Detection **EXTREMELY FAST & ACCURATE**...

Result

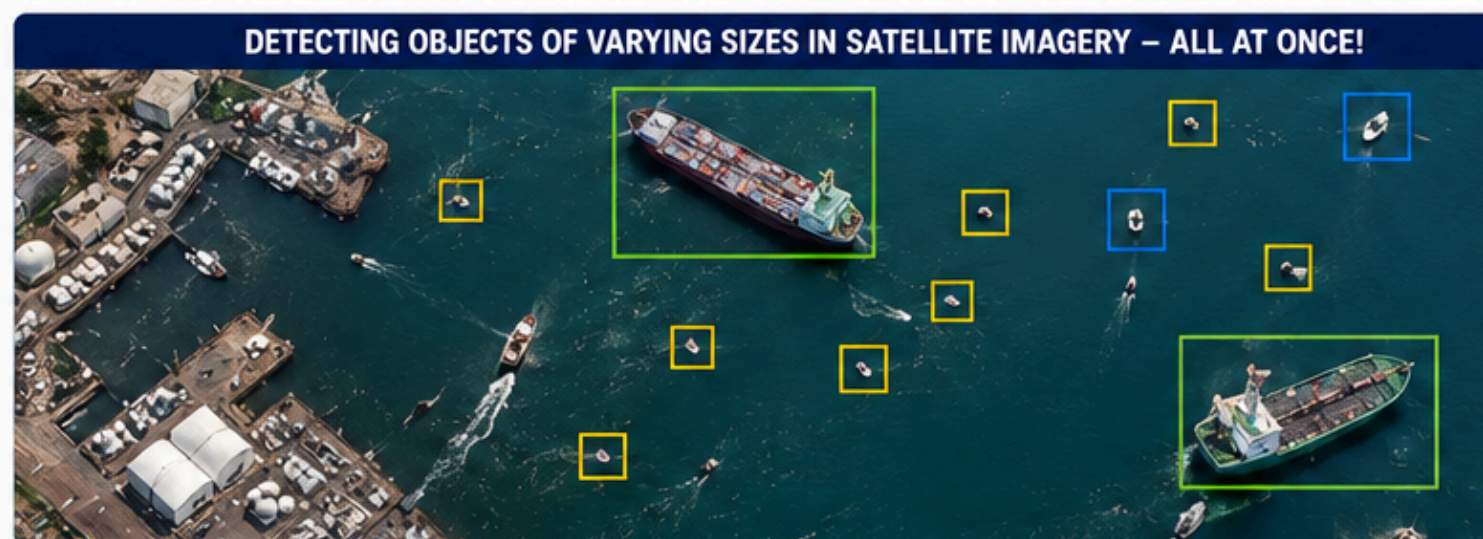
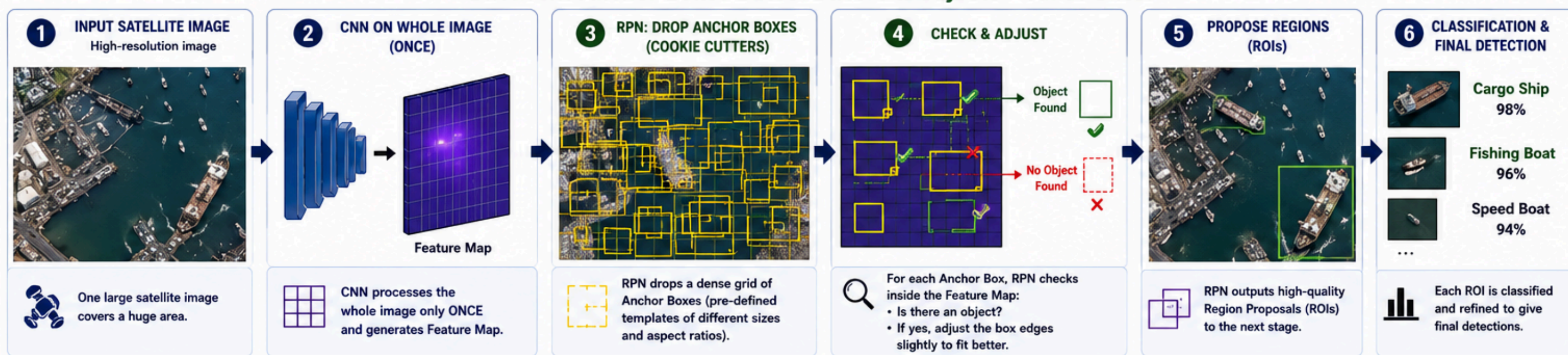
Perfect for Real-Time Applications like

"AQC"

~ 2 Sec per image → 7 FPS

2. Satellite Imagery Analysis → Maritime Surveillance

(Boats, Ships & *Illegal Fishing*)



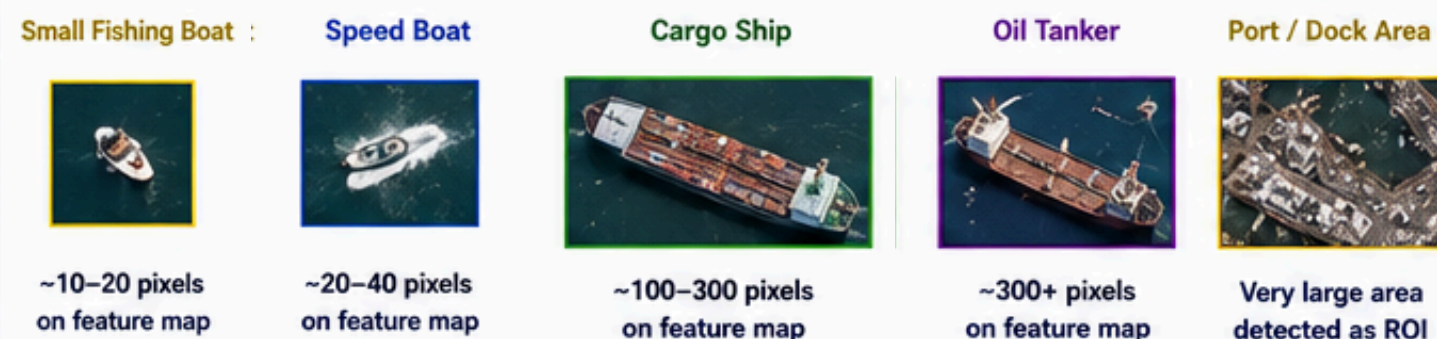
ANCHOR BOXES (TEMPLATES) USED IN RPN

- Very Small (e.g., small boats, buoys)
- Small (e.g., fishing boats)
- Medium (e.g., speed boats)
- Large (e.g., cargo ships)
- Extra Large / Wide (e.g., oil tankers, ports)
- ...

WHY THIS WORKS SO WELL?

- Pre-defined Anchor Boxes act like “cookie cutters” of different shapes and sizes.
- RPN performs fast template matching on the Feature Map (not on raw pixels).
- Everything runs on GPU pipeline – no complex search or calculations.
- Result: Lightning-fast detection of objects of all sizes in a single image.

FIND OBJECTS OF ALL SIZES – SMALL FISHING BOATS TO HUGE CARGO SHIPS



REAL-WORLD APPLICATIONS

- Satellite Imagery Analysis is critical for:
- Maritime Surveillance (boats, ships, illegal fishing)
 - Port Monitoring & Traffic Analysis
 - Disaster Response (damage assessment)
 - Border Security & Military Intelligence
 - Environmental Monitoring (oil spills, deforestation)
 - ... and many more!

Point 3

Limitations

Limitations in "7FPS" Processing System → Faster RCNN

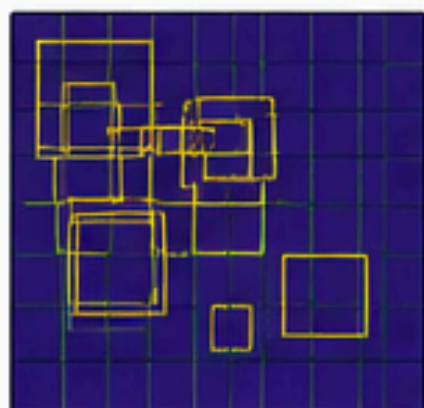
(For Real-Time **Self-Driving Cars**)

At 60 KMPH, even a small delay can be dangerous!

FASTER R-CNN: 2-STAGE PROCESS (7 FPS)

Stage 1: RPN (FIND)

Propose Regions (Anchor Boxes)



Stage 2: RCNN (CLASSIFY)

Classify Each Proposed Region



Car



Pedestrian



Traffic Light

Two steps = More computation = Slower speed

PARADIGM SHIFT IS NEEDED!

AI must see the entire scene at once, like humans do.

THE SPEED PROBLEM: REAL MATH

Car Speed: 60 KMPH

= **16.67 m/second**

At 7 FPS (7 updates per second):

Time between frames = $1/7 \text{ sec} \approx 0.143 \text{ sec}$

Distance traveled in one frame:

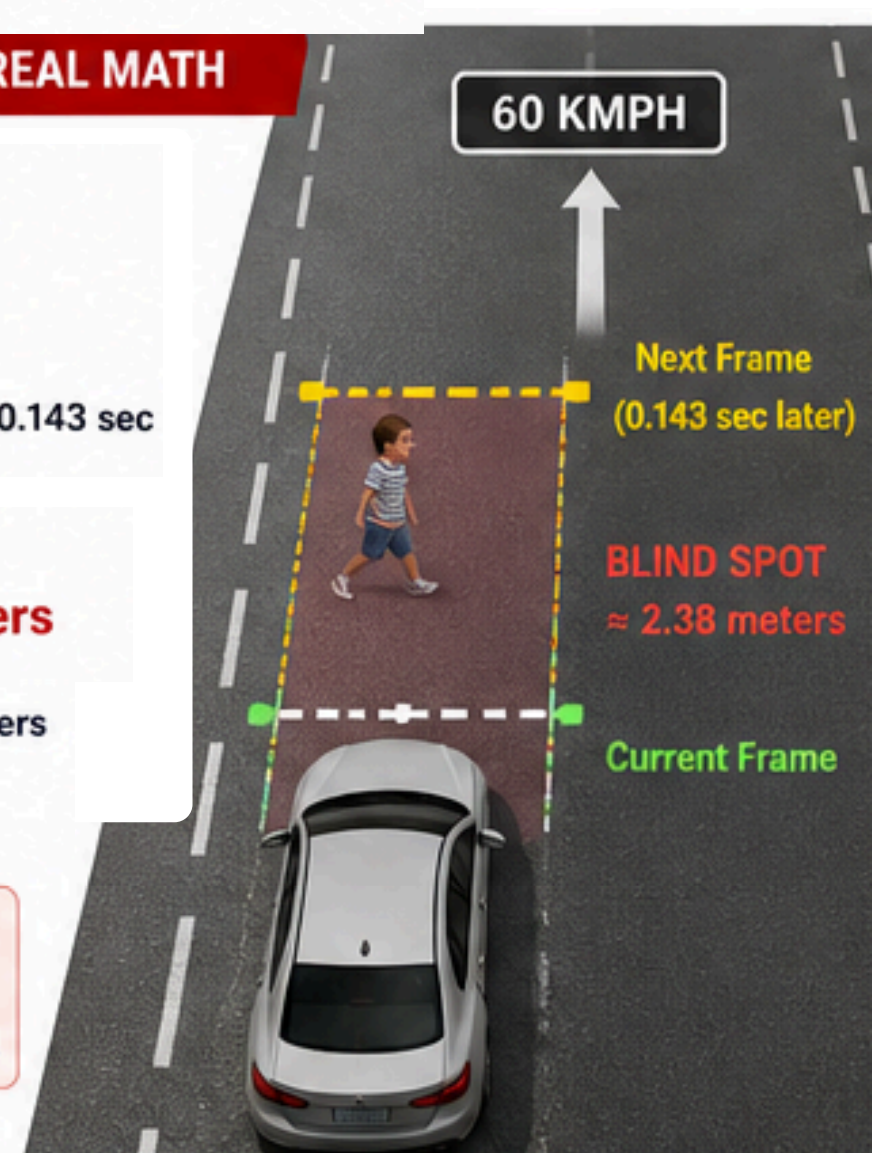
$16.67 \times 0.143 \approx 2.38 \text{ meters}$

So, the car goes BLIND for 2+ meters in every frame update!

! If a baby/pedestrian suddenly enters this blind spot, the car may not detect in time!



For Safe Driving, we need 30 FPS or 60 FPS (or more) — not 7 FPS!



WHY FASTER R-CNN CANNOT ACHIEVE REAL-TIME (30/60 FPS)



Two-stage pipeline (Find → Then Classify) = Inherently slower



Each stage adds computation & latency



Seeing small portions (boxes) one by one



Even with GPU optimization, Faster R-CNN ~7 FPS only

NOT ENOUGH for real-time autonomous driving!



Reality Check

Humans **don't first draw boxes** and then understand objects. We see the **whole scene at once**.

Point 4

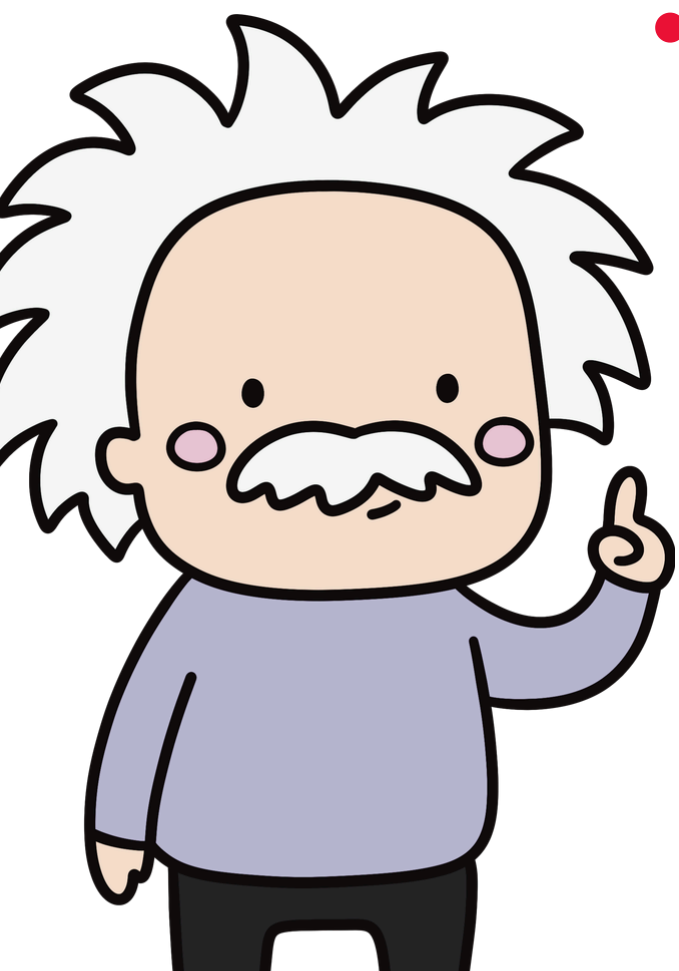
Takeaway

Key Takeaway Point

- **R-CNN** solved → **Object Localization**
- **Fast R-CNN** solved → **Repeated CNN computation**
- **Faster R-CNN** solved → **Proposal generation speed** (*Anchor boxes*)

But Still...

- Two-stage pipeline
- Find → Then classify
- Not truly real-time



Next question

"Can AI detect everything
in a single look"?

Yolo was born...

Point 9

Summary

Summary

1. Architectural Shift: **Faster RCNN**
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